Barbaros Zorlu

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EDUCATION

Virginia Tech **Blacksburg, VA**
B.S. Computer Science *Expected May 2027*

INDUSTRY EXPERIENCE

Microsoft **Chicago, IL**
AI/ML Intern *Jun 2026 – Aug 2026*

- Designed autonomous LLM agent workflows and deterministic tool-use layers in Azure for multi-step planning and execution tasks.
- Built an automated evaluation pipeline to benchmark agent outputs across quality, latency, reliability, and security constraints.

Uplifty AI **Chicago, IL**
Full Stack Developer Intern *Jun 2026 – Aug 2026*

- Architected a two-layer caching system for React Native to mitigate production memory spikes and improve asset resolution across mobile views.
- Refactored shared UI patterns across 40+ React Native screens into reusable components, eliminating duplicated implementations and improving maintainability.

Google **Chicago, IL**
Product Management Intern *Jul 2025 – Aug 2025*

- Built an end-to-end Gmail email classifier using Python, TF-IDF, and logistic regression to organize messages into personalized smart categories.
- Integrated the ML pipeline into a Streamlit-based Gmail prototype supporting personalized folders and automated email categorization.

Manufacturing Renaissance **Chicago, IL**
Software Engineering Intern *Jun 2025 – Jul 2025*

- Developed a SwiftUI application connecting Chicago residents with 500+ manufacturers through a searchable directory, job board, and contact system.
- Integrated dynamic manufacturer data into reusable UI components, replacing manual content updates across multiple application screens.

RESEARCH EXPERIENCE

JudgeJack — NeurIPS 2026 Workshop **Remote**
Researcher & Co-author *May 2026 – Sep 2026*

- Co-authored research on a data-poisoning backdoor that manipulates LLM verifier decisions and test-time compute.
- Built a matched-pair evaluation pipeline with Qwen2.5-1.5B-Instruct across 21,334 holdout examples to isolate the trigger's causal effect.

Virginia Tech Department of Computer Science **Blacksburg, VA**
LLM Systems Undergraduate Researcher *Jan 2026 – May 2026*

- Engineered a multi-stage LLM pipeline to extract and generalize math problems from HTML/PDF sources into structured templates with database-backed storage.
- Designed a hybrid duplicate-detection pipeline combining metadata pre-filtering with LLM semantic verification to reduce redundant model comparisons.

Virginia Tech College of Engineering **Blacksburg, VA**
NLP Undergraduate Researcher *Aug 2025 – May 2026*

- Built an NLP pipeline over 1,109 student-feedback samples using embeddings, UMAP, and HDBSCAN to transform unstructured responses into interpretable semantic themes.
- Refactored the application into modular data, modeling, and service layers and integrated clustering outputs into an interactive Streamlit dashboard.

LEADERSHIP & HONORS

Association for Computing Machinery (ACM), Wilbur Wright College **Chicago, IL**
President *May 2024 – May 2025*

- Led 8 officers and 100+ members in organizing technical workshops, hackathons, interview-prep programs, and industry events.

Honors: Outstanding President of a Student Organization; Two-Time National Triathlon Champion

SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript, SQL, Swift
Technologies & ML: Azure, React, React Native, Node.js, Docker, Git/GitHub, Streamlit, SwiftUI, scikit-learn, pandas, UMAP, HDBSCAN